next generation of UAVs toward higher endurance, reliability, and intelligence (Choi, et al. 2023).

This chapter provides a comprehensive overview of drone technology, tracing its historical evolution, classification, design elements, and operational principles. It also highlights key enabling technologies and emerging trends that continue to shape the future of UAVs as indispensable tools in defense, industry, and daily life.

1.2 The Impact of Drone Technology

Drone technology has emerged as a transformative force with far-reaching implications across military, commercial, and civilian domains. By combining advancements in aerodynamics, electronics, communication systems, and artificial intelligence, drones have evolved from simple remote-controlled aircraft into intelligent, autonomous platforms capable of performing complex tasks with precision, speed, and reliability (Singhal, 2021). Their growing influence signifies not just technological progress but also a structural shift in how industries operate, governments manage security, and societies engage with the aerial dimension of modern innovation.

1.2.1 Drones in military applications

Drones have profoundly transformed military strategy and defense operations by introducing a new dimension of intelligence, surveillance, and reconnaissance (ISR) capabilities. Modern militaries rely heavily on UAVs for missions that require persistence, precision, and safety in high-risk environments. Drones such as the *Predator*, *Global Hawk*, and *Heron* are equipped with high-resolution cameras, thermal imaging sensors, radar systems, and real-time communication links that provide commanders with continuous situational awareness.